

CAPSTONE PROJECT SYNOPSIS
ON
“HOSTEL LEAVE MANAGEMENT SYSTEM”

In partial fulfilment for the award of the degree of
Bachelor of Engineering
IN
(Computer Science & Engineering (AI &DS))



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Title of the Project

Hostel leave management system

Introduction:

In many hostels, the leave process is still done manually using paper forms and registers. This method takes more time and can cause mistakes, loss of records, and confusion for both students and hostel authorities. Students have to fill out forms, and wardens need to check and approve them manually.

The Hostel Leave Management System is a digital system that helps to manage hostel leave in an easy and organized way. Through this system, students can apply for leave online, and wardens can approve or reject the request. All leave records are stored safely in one place, which makes tracking and managing student leaves simple and efficient.

The system provides a centralized platform that allows authorized users to manage hostel leave requests easily and efficiently.

Problem Statement :

Manual hostel leave management suffers from several issues such as:

- Hostel leave management is mostly done using manual paper-based systems.
- Students must physically visit multiple authorities to apply for leave.
- The manual process is time-consuming and inefficient.
- Chances of data manipulation or loss
- Difficulty in tracking student movement
- No centralized database for leave history
- Maintaining separate registers for leave, mess-off, and permanent leave increases workload.

Objective of the System :

The main objective of hostel leave management system are :

- To digitize the hostel leave application process
- To reduce manual work and errors
- To provide leave approval and status updates
- To maintain a centralized database of student leave records
- To improve hostel security and administration efficiency
- To maintain accurate and organized leave records for hostel administration
- To save time and effort for both students and hostel staff.

Proposed System:

The proposed system is a web/mobile-based application where:

- Students can apply for leave online
- Wardens/authorities can approve or reject leave requests
- Leave records are stored securely in a database
- Notifications are sent to students regarding leave status
- Reduces paperwork and manual effort

System Features :

- Student login and profile management
- Online leave application form
- Warden/admin approval panel
- Leave history and reports
- Notification system (SMS/Email/App alerts)
- Secure authentication and role-based access

Methodology :

First, the current hostel leave process is studied. Then an online system is created to manage leave applications and approvals. Students apply for leave through the system, and wardens approve or reject the requests. The system is checked properly and then used for daily hostel work. This system helps save time, reduces errors .

Tools and Technologies :

Front end : HTML5, CSS3, Bootstrap

Back end : Python, Flask, Flask-SQLAlchemy, Flask-WTF, AI- ML

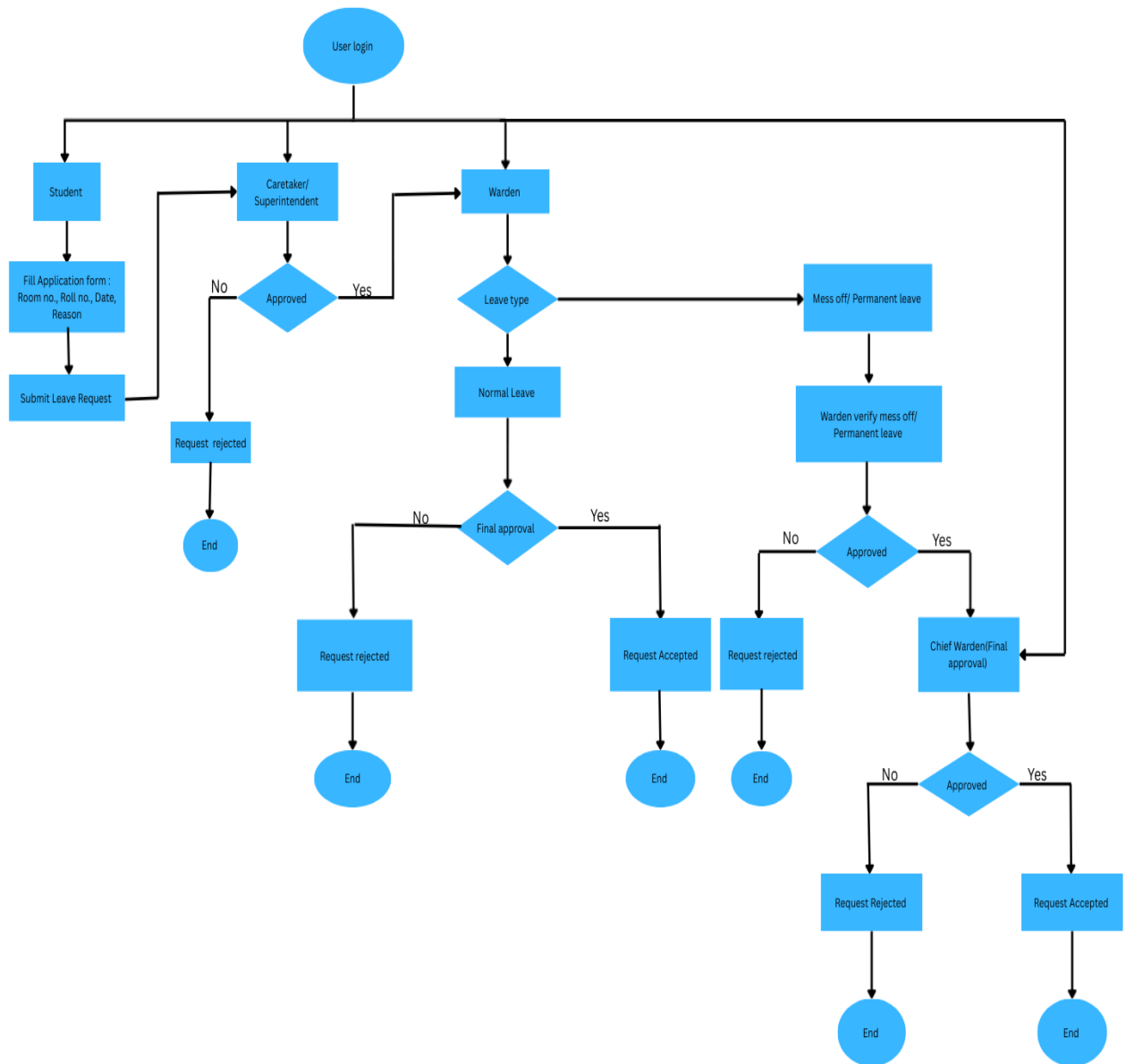
Integration Database : SQLite (with SQLAlchemy)

Development tools : Visual Studio Code (VS Code), Pipenv / Virtualenv, Postman

Expected Outcomes :

- Faster and smoother hostel leave application process
- Reduction in paperwork and manual record keeping
- Accurate and secure storage of student leave data
- Easy tracking of leave status for students and wardens
- Time-saving for both students and hostel staff

Project Design :



Conclusion:

The Hostel Leave Management System provides an effective and reliable solution for managing student leave in hostels. By replacing the manual process with a digital system, it reduces paperwork, saves time, and improves accuracy. The system ensures transparency in leave approval and allows easy access to leave records for both students and hostel authorities. Overall, this system enhances hostel administration and provides a more organized and efficient leave management process.